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CSA1717 – ARTIFICIAL INTELLIGENCE

1. Print a message

```
>>> print("welcome to artificial intelligence")
welcome to artificial intelligence
```

2. Addition of 2 numbers

```
>>> #addition of 2 numbers
>>> #input
>>> a=6
>>> b=7
>>> c=a+b
>>>
>>> #output
>>> print("the sum of a and b is:",c)
the sum of a and b is: 13
>>>
```

3. subtraction of 2 numbers

```
>>> #subtraction of 2 numbers
>>> #input
>>> a=15
>>> b=6
>>> c=a-b
>>>
>>> #output
>>> print("the subtraction of a and b is:",c)
the subtraction of a and b is: 9
```

4. Multiplication of 2 numbers

```
>>> #multiplication of 2 numbers
>>> #input
>>> a=3
>>> b=9
>>> c=a*b
>>>
>>> #output
>>> print("the multiplication of a and b is:",c)
the multiplication of a and b is: 27
```

5. Division of 2 numbers

```
>>> #Division of 2 numbers
>>> #input
>>> a=4
>>> b=24
>>> c=a/b
>>>
>>> #output
>>> print("the division of a and b is:",c)
the division of a and b is: 0.16666666666666666
```

6. Swapping of 2 numbers

```
>>> #swapping of 2 numbers
>>> a=5
>>> b=14
>>> print("Before:", a, b)
Before: 5 14
>>> a, b = b, a
>>> print("After:", a, b)
After: 14 5
```

7. Circulate the values of n variables

```
>>> #Circulate the values of n variables
>>> a, b, c, d = 1, 2, 3, 4
>>> print("Before:", a, b, c, d)
Before: 1 2 3 4
>>> a, b, c, d = d, c, a, b
>>> print("After:", a, b, c, d)
After: 4 3 1 2
>>>
```

8. Distance between two points

```
>>> #Distance between two points
>>> import math
>>> x1, y1 = 12, 3
>>> x2, y2 = 3, 4
>>> distance = math.sqrt((x2 - x1)**2 + (y2 - y1)**2)
>>> print("Distance:", distance)
Distance: 9.055385138137417
```

9. Conditional (if)

```
>>> #Conditional (if)
>>> age=18
>>> if age>=18:
...     print("You are eligible to vote.")
...
...
You are eligible to vote.
```

10. Alternative (if-else)

```
>>> #Alternative (if-else)
>>> num = 6
>>> if num % 2 == 0:
...     print("Even")
... else:
...     print("Odd")
...
...
Even
```